

- (d) the water level will not change
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (d)

See the explanation of above question.

51. A body floats at 4°C water. If temperature reaches 100°C then –

- (a) The body will sink
(b) No change
(c) More part of the body will submerge
(d) The body will float freely

M.P.P.C.S. (Pre) 1999

Ans. (a)

When the water is heated, its density decreases. The density of the water at 100°C is at the lowest, so the body will sink. Water density is maximum at 4°C, so the body floats.

52. Why it is easier to swim in the sea than in a river?

- (a) The sea water is salty
(b) The sea water is deep
(c) The sea water is heavier
(d) The sea water is lighter

U.P. Lower Sub. (Pre) 2002

Ans. (a)

It is easier to swim in sea water than in a river because, the sea water contains salts which increase the density of water and also increase its upthrust so, the chances of sinking get reduced and one can easily swim in such water.

53. A body-weight is highest in which among the following?

- (a) Air (b) Water
(c) Hydrogen (d) Nitrogen

U.P. P.C.S. (Mains) 2017

Ans. (c)

The upthrust force or buoyant force in Hydrogen will be less than other matters because it is the lightest gas. So, option (c) is the correct answer.

54. When a ship enters a sea from a river, what happens?

- (a) It rises a little
(b) It remains at the same level
(c) It sinks a little
(d) It immersed in the bottom of the sea.

U.P. Lower Sub. (Pre) 2004

U.P. P.C.S. (Pre) 1992

Ans. (a)

When a ship enters a sea from a river, the ship is elevated due to the higher density of salty sea water than the river.

55. Assertion (A) : When a ship enters a sea from a river, it rises a little.

Reason (R) : The density of sea water is greater than that of river water.

In the context of the above statements, which of the following is true :

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

U.P. P.C.S. (Pre) 2003

U.P.U.D.A./L.D.A. (Pre) 2002

Ans. (a)

See the explanation of above question.

Optics

Notes

- Light is an electromagnetic radiation within a certain portion of electromagnetic spectrum. It lies between the infrared (with longer wavelengths) and the ultraviolet (with shorter wavelengths).
- Its wavelength ranges from $3900 \text{ \AA}$ to $7800 \text{ \AA}$ or about 4×10^{-7} to 7×10^{-7} metre.
- It is invisible in itself but helps in visibility of matter.
- It travels in a straight line.
- It can travel through vacuum (sound waves cannot do so).
- Light has dual nature :
(a) **Particle nature** – behaves in the form of particles.
(b) **Wave nature** – behaves in the form of waves.

Speed of light in different mediums

Sl.No.	Medium	Speed of light in m/s
1.	Vacuum	3×10^8
2.	Water	2.25×10^8
3.	Glass	2×10^8
4.	Diamond	1.24×10^8

- Above chart shows that light speed differs in different mediums.
- Light speed is maximum in vacuum.

Refractive Index :

- Light speed is less in a medium with the high refractive index.

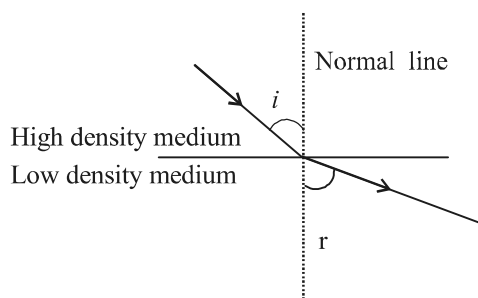
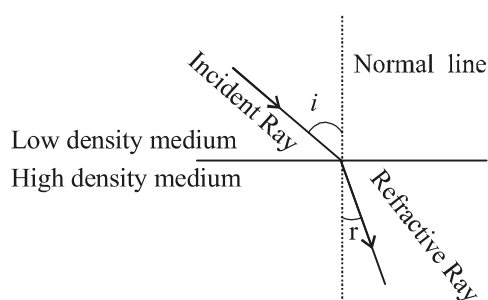
- If the refractive index of a medium is μ and light speed is u and c in that medium and vacuum respectively then

$$\mu = \frac{c}{u}$$

- Thus, the ratio of the velocity of light in a vacuum to its velocity in a specified medium is known as the refractive index of that medium.
- (a) Refractive index of water -1.33
(b) Refractive index of glass-1.50
- It is represented by Greek letter - (μ)
 - Since the light speed is maximum in vacuum, so the value of the refractive index of any medium is always more than 1 in relation to vacuum.
 - Refractive index is also called optical density.
 - Minimum the speed of light in a material, maximum is its optical density.

Refraction of Light :

- Light rays travel in a straight line in a homogenous medium.
- When light rays enter from one transparent medium to another transparent medium it deviates from its original path.
- This phenomenon is called refraction.
- The light rays passing in the first medium is called incident rays.
- Light rays which enter into next medium is called refractive rays.
- Incident point is the point at which a light ray enters in an optical system.
- The **angle of incidence** is the angle between a ray incident on a surface and the line perpendicular to the surface at the point of incidence, called the normal.
- Angle of refraction** is the angle between the normal line and refractive ray on the surface.
- By experiments it has been observed that when light ray enters from low-density medium to higher density medium it bends towards the normal line while entering from higher density medium to low-density medium it goes far from the normal line.

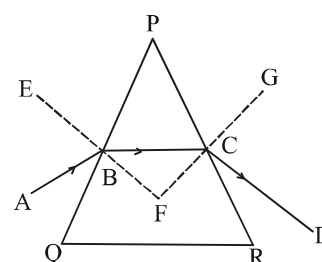


Examples of Refractive Phenomenon :

- An immersed rod in water seems small and bent at the surface when it is observed from outside.
- A coin within a pot with water seems above from its original position.
- Twinkling of stars.

Refraction of light in a Prism :

- Prism is a homogenous transparent medium surrounded by plane surface at an acute angle.
- Generally it is made up of glass.

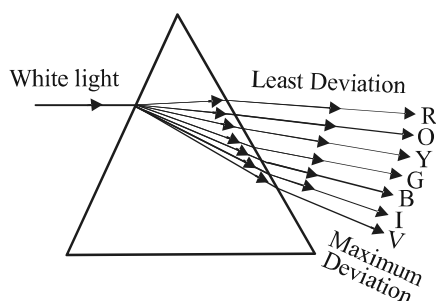


- In the above figure, light ray AB after falling on refractive surface PQ of the prism, refracted in BC direction bending towards normal line EBF. Refracted ray BC on surface emerges out in the air in CD direction going away from normal line GCF. Thus, ABCD is the path of light in Prism.

Dispersion of Light :

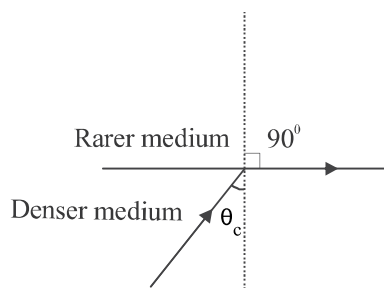
- White light of the sun is a mixture of seven (7) different colors.
- These are violet, indigo, blue, green, yellow, orange & red (**VIBGYOR**).
- The separation of visible light into its different colors is known as **dispersion**.
- In spectrum violet color is at lower position and the red color is at the upper position.
- It shows that violet ray dispersed maximum and dispersion of red ray is minimum.
- Speed of different colors of light are same in vacuum and in the air but differs in matters.
- Thus the refractive index of any matter differs for different colors of light.
- Speed of violet light is minimum and speed of red light is maximum in the glass.

- So the refractive index of glass is maximum for violet light and minimum for red light and the violet light deviates most and red light deviates least.
- Violet ray is with lowest wavelength and highest frequency while red ray is with highest wavelength and lowest frequency.

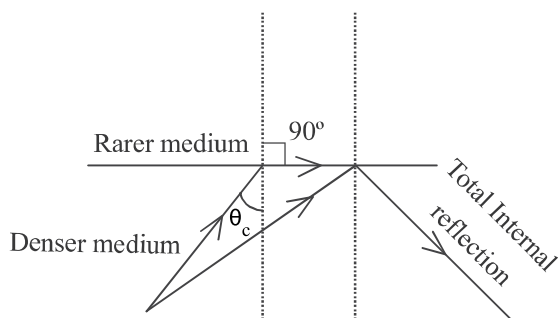


Critical Angle and Total Internal Reflection :

- Critical angle (θ_c) is the angle of incidence beyond which rays of light passing through a denser medium to the surface of a less dense medium are no longer refracted but totally reflected.
- Its value depends upon the nature of both mediums and color of rays.
- For critical angle light must pass from a denser medium to rarer medium.
- For glass-air surface the value of the critical angle is approximately 42° and for the water-air surface it is 48.6° .



- The complete reflection of light ray reaching an interface with a less dense medium when the angle of incidence exceeds the critical angle is known as **total internal reflection**.



- Mirage formation is a result of the refraction and the total internal reflection of light in the air.
- Optical fibres and Endoscope function on the principle of total internal reflection.

Reflection :

- When light falls on the surface it may be absorbed, transmitted or reflected.
- Reflection is when light bounces off an object.
- Matters with polished and shining surface reflects more light than the rough surface of the matter.
- Silver is the better reflector of light.

Mirror :

- A smooth well polished reflecting surface is called the mirror.
- Mirrors are of two types-
(i) Plane Mirror
(ii) Spherical Mirror

Plane Mirror –

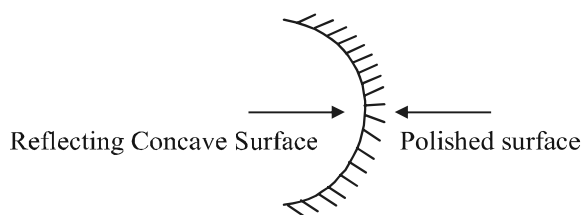
- A plane mirror is a thin, flat and smooth sheet of glass with a shining layer of silver on one side.
- The mirror used to see the face is a plane mirror.
- Our face image formed in the plane mirror can not be obtained on screen, it can only be observed in the mirror.
- So, the image formed in a plane mirror is an example of **virtual image**.
- Such an image is also known as imaginary or unreal image.
- The size of the image formed in the mirror is just equal to the size of the object.
- The image and objects are at the same distance from the mirror at opposite sides.
- The image is a laterally inverted mirror image of the object. If a person is reflected in a plane mirror, the image of his right hand appears to be the left hand of the image.

Spherical Mirror –

- A spherical mirror is a mirror which has the shape of a piece cut out of a spherical surface.
- These are of two types –
(i) Concave Mirror
(ii) Convex Mirror

Concave Mirror –

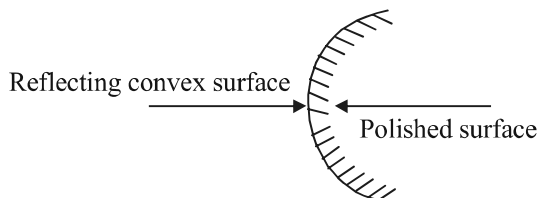
- Concave mirrors have inwards curved reflecting surface.



- Dentists use concave mirror for the examination of patient's teeth producing big images of teeth.
- Concave mirror are used as a reflector, for obtaining a sharp beam of light in torches, headlight of vehicles and searchlights.
- Concave mirrors are also used for shaving to get clear image of face, etc.

Convex Mirror –

- Convex mirrors have outward curved reflective surface.



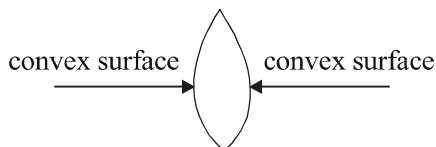
- It is used as rear - view side mirrors in vehicles.

Lens :

- Lens is a **homogenous** medium surrounded by two curved surfaces or one curved and another plane surface.
- Curved surface may be spherical, cylindrical or elliptical, but these are mostly spherical.
- Lens is of two types-
(i) Convex Lens
(ii) Concave Lens

Convex Lens –

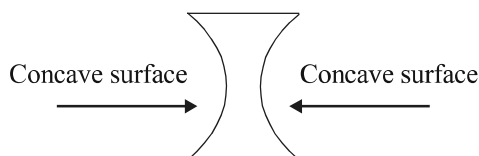
- Convex lens is thick in center but thin at corner.



- Convex lens is also known as a converging lens because it converges the parallel light rays at one point. Its focal length is positive. Objects appear closer and larger and their images are real and inverted through a convex lens.
- Convex lens is used to correct far-sightedness/hypermetropia, an eye defect in which near objects are not clearly visible but long-distance objects are clearly visible.

Concave Lens –

- Concave lens is thin at the centre but thick at the corner.



- It is also known as a divergent lens because it diverges the parallel light rays. Its focal length is negative. Objects appear smaller and farther through the concave lens and their images are virtual.

- Concave lens is used to correct near-sightedness/ myopia, an eye defect in which far objects are not clearly visible but near objects are clearly visible.

Scattering :

- Scattering is the phenomenon by which a beam of light is redirected in many different directions when it interacts with a particle of matter.
- Scattering occurs when a particle of light is fully absorbed and then emitted, while reflection is when a wave/particle is simply reflected off the surface without interacting.
- Light of shorter wavelength is scattered much more than the light of longer wavelength.
- Color of sky appears to blue due to the scattering of light.

Question Bank

1. Which of the following radiations has the highest energy per quantum?

- (a) 320-400 nm (b) 200-280 nm
(c) 280-320 nm (d) 400-600 nm

Uttarakhand P.C.S. (Pre) 2012

Ans. (b)

Photon energy is directly proportional to the wave frequency and inversely proportional to the wavelength.

$$E = \frac{hc}{\lambda} \quad \text{where, } E = \text{energy of photon}$$

h = Planck's constant

c = speed of light in a vacuum

λ = wavelength

Thus, among the given options 200-280 nm radiation has the highest energy per quantum. Ultraviolet radiation contains sufficient energy per quantum to damage living cells. The ultraviolet spectrum is divided into UV-A (320-400 nm) responsible for tanning the skin; UV-B (280-320 nm) responsible for skin cancer and vitamin D synthesis; and UV-C (below 280 nm), potentially the most harmful waveband but absorbed almost completely by molecular oxygen in the stratosphere.

2. The visible range of solar radiation is –

- (a) 100-400 nm (b) 400-700 nm
(c) 740-10000 nm (d) None of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

The electromagnetic spectrum encompasses all type of radiations. The part of the spectrum that reaches earth from the sun is between 100 nm to 10^6 nm. This band is broken into three ranges - Infrared (above 700 nm), Visible (400 to 700 nm), Ultraviolet (below 400 nm).

3. The wave length extension of visible light is –

- (a) Between 200-900 nm (b) Between 250-850 nm
(c) Between 300-800 nm (d) Between 390-780 nm

U.P.P.C.S.(Mains) 2005

Ans. (d)

Visible light is that part of electromagnetic radiation which can be seen by human eyes. Visible light is usually defined as having wavelengths in the range of 400-700 nm (4000-7000 Å), between the infrared and ultraviolet. A typical human eye will respond to wavelengths from about 380 to 750 nm. Thus, among the given options, option (d) will be the right answer.

4. The wavelength of visible spectrum is in the range :

- (a) 1300 Å - 3900 Å (b) 3900 Å - 7600 Å
(c) 7800 Å - 8200 Å (d) 8500 Å - 9800 Å
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

See the explanation of above question.

5. Human eye is most sensitive to visible light of the wavelength :

- (a) 6050 Å (b) 5500 Å
(c) 4500 Å (d) 7500 Å

U.P.P.S.C. (R.I.) 2014

Ans. (b)

Eye is most sensitive to yellowish-green light that is light of wavelength about 5550 Å or 555 nm.

6. Among the colours mentioned below, the eye is most sensitive to colours between :

- (a) Red and green (b) Orange and yellow
(c) Green and blue (d) Yellow and green

U.P.P.C.S. (Mains) 2008

Ans. (d)

In normal lighting conditions, photo receptor cells in human eye are most sensitive to colours between the wavelengths of 530-570 nanometre. The sensitivity of our eyes is most for yellowish-green colour.

7. Which of the following statement is not correct?

- (a) Human ear becomes most sensitive at 1000 Hz.
(b) Human eye becomes most sensitive for red colour.
(c) Persistence of hearing is 1/10 second.
(d) Persistence of vision is 1/16 second.

U.P. R.O./A.R.O. (Pre) 2021

Ans. (b)

The human ear is most sensitive to sounds between 1000 and 5000 Hz. The human eye is most sensitive to yellowish- green colour. When fully light-adapted, the human eye features a wavelength response from around 400 to 700

nanometers, with a peak sensitivity at 555 nanometers (in the yellowish-green region of the visible light spectrum). The impression of sound heard by our ears persist in our mind for 1/10th of a second. If another sound is heard before 1/10 second is over, the impression of the two sound mixes up and our mind cannot differentiate among the two. Persistence of vision is a phenomenon of the human eye wherein the image formed on the retina is retained by the brain for 1/16th of a second. Hence, statement of option (b) is incorrect.

8. The nature of radiation of light is-

- (a) Like wave
(b) Like particle
(c) Like both of wave and particle
(d) Like neither of wave nor of particle

42nd B.P.S.C. (Pre) 1997-98

Ans. (c)

The light has a dual nature, sometimes it behaves as wave and sometimes as particle. In the later part of the 19th century and in the beginning of 20th century, it was realized that black body radiation and the photoelectric effect can be understood only on the basis of particle model of light. Some experiments require light to be a wave, while others require light to be a particle. This led to the acceptance of dual nature of light. Quantum mechanics explains the duality of light by describing it as a wave-packet. A wave-packet refers to waves that may interact either as spatially localized, acting as particle, or interacting like waves.

9. Light appears to travel in a straight line because –

- (a) It consists of small particles.
(b) The velocity of light is very large.
(c) The wavelength of light is very small.
(d) Light is reflected by the surroundings.

U.P.P.S.C. (R.I.) 2014

Ans. (c)

Due to its very small wavelength, the light appears to travel in a straight line.

10. Which of the following statement is true?

- (a) The light has a greater velocity.
(b) The sound has the greatest velocity.
(c) Celestial objects have the greatest velocity.
(d) Rocket has the greatest velocity.

U.P. Lower Sub. (Pre) 2002

Ans. (a)

The speed of light in a vacuum is 3.00×10^8 m/s, while the speed of sound in vacuum is zero and in air is 343 m/sec. The speed of light in glass is 2.0×10^8 m/s. The velocity of light always greater than the velocity of sound. The velocity of celestial objects and the rockets is quite low than the velocity of light.

- 11. The speed of light in vacuum is nearly –**
 (a) 3×10^{10} metre/sec (b) 3×10^8 metre/sec
 (c) 3×10^8 km/sec (d) 3×10^8 light years
43rd B.P.S.C. (Pre) 1999

Ans. (b)

See the explanation of the above question.

- 12. The speed of light is –**
 (a) 9×10^2 m/sec (b) 3×10^{11} m/sec
 (c) 3×10^8 m/sec (d) 2×10^4 m/sec
Jharkhand P.C.S. (Pre) 2003

Ans. (c)

See the explanation of above question.

- 13. The speed of light in air depends on :**
 (a) Pressure (b) Density
 (c) It is independent of pressure, temperature and density
 (d) Temperature
Jharkhand P.C.S. (Pre) 2021

Ans. (*)

The speed of light depends on the optical density of the (transparent) medium in which it is travelling. The optical density of a medium is not the same as its physical density (which refers to the mass/volume ratio). The refractive index is the indicator of optical density of the medium. The optical density of air depends on its temperature, pressure and humidity. Hence, the speed of light in air depends on its optical density, temperature and pressure. Atmospheric refraction (e.g. **mirage**) happens due to change in these gradients.

- 14. Velocity of light is maximum in –**
 (a) Diamond (b) Water
 (c) Vacuum (d) Hydrogen
U.P.P.S.C. (R.I.) 2014

Ans. (c)

The speed of light in a vacuum is maximum ($c = 3 \times 10^8 \text{ ms}^{-1}$). The speed of light decreases gradually in optically denser mediums like water or glass. It is least in diamond.

- 15. Light travels at a maximum speed in :**
 (a) Air (b) Water
 (c) Vacuum (d) None of these
Uttarakhand P.C.S. (Pre) 2021

Ans. (c)

See the explanation of above question.

- 16. The speed of light will be minimum while passing through–**
 (a) Glass (b) Vacuum
 (c) Water (d) Air
I.A.S. (Pre) 1994
67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

Any of the object or material which has the highest refractive index has the minimum speed of light. The glass has the highest refractive index among the given options. Hence, among the given options, the speed of light will be minimum while passing through glass.

- 17. The time taken to reach the sunlight up to the surface of Earth is approximately –**
 (a) 4.2 sec (b) 4.8 sec
 (c) 8.5 minutes (d) 3.6 hrs.
42nd B.P.S.C. (Pre) 1997

Ans. (c)

The sunlight takes about 499 second or about 8.3 minutes to reach to the Earth. Thus, option (c) is the right answer.

- 18. Light from the Sun reaches Earth in about :**
 (a) 2 minutes (b) 4 minutes
 (c) 8 minutes (d) 16 minutes
U.P.P.C.S. (Mains) 2006

Ans. (c)

See the explanation of above question.

- 19. The sunlight from the Sun to the Earth reaches in :**
 (a) 5 minutes (b) 6 minutes
 (c) 8 minutes (d) 10 minutes
 (e) None of the above/More than one of the above
65th B.P.S.C. (Pre) 2019

Ans. (c)

See the explanation of above question.

- 20. Which of the following has the highest velocity?**
 (a) Cosmic rays (b) Light
 (c) Electron (d) Supersonic wave
U.P. P.C.S. (Pre) 2018

Ans. (b)

Nothing can travel faster than the speed of light. The light waves can travel through a vacuum. The sound waves can not travel through a vacuum. The speed of light is maximum in a vacuum. Most galactic cosmic rays have energies between 100 MeV (corresponding to velocity for protons of 43% of the speed of light) and 10 GeV (corresponding to 99.6% of the speed of light). The speed of highest energy cosmic rays could be extremely close to but still below, the speed of light in a vacuum.

- 21. Which of the following does not change when light travels from one medium to another?**
 (a) Velocity (b) Wavelength
 (c) Frequency (d) Refractive index
 (e) None of the above / More than one of the above
66th B.P.S.C. (Pre) 2020

Ans. (c)

When light travels from one medium to another, the frequency of the light does not change, while its velocity, wavelength and refractive index of the medium are changed.

22. When light waves pass from air to glass the variable affected are –

- (a) Wavelength, frequency and velocity
- (b) Velocity and frequency
- (c) Wavelength and frequency
- (d) Wavelength and velocity

I.A.S. (Pre) 2001

Ans. (d)

See the explanation of above question.

23. When light passes from one medium to another medium, which one of the followings does not change :

- (a) Intensity
- (b) Velocity
- (c) Wavelength
- (d) Frequency

U.P.P.C.S. (Mains) 2003

Ans. (d)

See the explanation of above question.

24. When a ray of light travels from a denser medium to a rarer medium then

- (a) It speeds up and bends away from the normal
- (b) It speeds up and bends towards the normal
- (c) It slows down and bends away from the normal
- (d) It slows down and bends towards the normal

Chattisgarh P.C.S. (Pre) 2018

Ans. (a)

When a ray of light travels from a denser medium to a rarer medium it speeds up and bends away from the normal. If ray of light travels from a rarer to a denser medium, it slows down and bends towards the normal in the denser medium.

25. The formation of glittering colours in thin foam of soap is the result of which of the following phenomenon –

- (a) Total reflection and interference
- (b) Total refraction and dispersion
- (c) Diffraction and dispersion
- (d) Polarization and interference

I.A.S. (Pre) 1993

Ans. (a)

The formation of glittering colour in thin foam of soap is the result of the phenomenon of total reflection and interference. When light waves hit a bubble, some of them bounce straight back off the outer part of the soap film. Others refracted on through but then bounce off the inner part of the film. Now interference of refracted and reflected light gives soap foam glittering colour.

26. Total internal reflection can take place when light travels from :

- (a) Diamond to glass
- (b) Water to glass

(c) Air to water

(d) Air to glass

I.A.S. (Pre) 1996

Ans. (a)

Total internal reflection takes place when a ray of light is travelling from denser to rarer medium and angle of incidence is greater than the critical angle, then the ray is completely reflected from the surface. So according to options, the situation of option (a) is correct where total internal reflection takes place, it is because diamond is denser than glass and it has very high refractive index.

27. A cut diamond shines brilliantly due to

- (a) Some inherent property
- (b) Total internal reflection
- (c) Its molecular structure
- (d) Absorption of light

U.P.P.C.S. (Pre) 2017

U.P.P.C.S. (Mains) 2015

Ans. (b)

A cut diamond shines brilliantly due to total internal reflection of light. Diamond shines more than a glass piece because of its low critical angle of approx. 24.4 degrees. This is the reason why diamond shines more than a glass piece of same shape.

28. The diamond appears lustrous because of :

- (a) Reflection
- (b) Refraction
- (c) Total Internal Reflection
- (d) Scattering

U.P.P.C.S. (Mains) 2012

U.P.P.C.S. (Pre) 1996

Ans. (c)

The refractive index of diamond is very high and therefore its critical angle is very low. Diamond achieves brilliance from total internal refraction. It has been cut or designed in such a way that if light enters into it, incident light strikes many of the internal surfaces. After many such total internal reflections, the colours in the light are separated and hence seen individually and therefore diamond has extra ordinary shining.

29. What is the reason behind the extra ordinary shining of the diamond cuts in a proper way?

- (a) Because it has a higher transparency
- (b) Because it has a higher refractive index
- (c) Because it is so hard
- (d) Because it has a fixed deviation floors

I.A.S. (Pre) 1994

Ans. (b)

See the explanation of above question.

30. Given below are two statements, one labeled as Assertion (A) and other labeled as Reason (R) :

Assertion (A) : A diamond sparkles more than a glass imitation cut to the same shape.

Reason (R) : The refractive index of diamond is less than that of glass.

In the context of the above two statements, which one of the following is correct ?

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not a correct explanation of A
- (c) A is true but R is false.
- (d) A is false but R is true.

I.A.S. (Pre) 1995

Ans. (c)

A diamond sparkles more than a glass crystal of the same shape and size, because the refractive index of diamond is very high ($n_d = 2.42$). Therefore its critical angle get reduced which results in less opportunity for light to exit. So the light reflects off the diamond's inner surfaces several times before it finally exits. The refractive index of glass $n_g = 1.51$. Thus, statement A is true but reason R is false.

31. Optical fibre works on the principle of :

- (a) Total internal reflection
- (b) Refraction
- (c) Scattering
- (d) Interference

U.P. U.D.A./L.D.A. (Pre) 2010

U.P.P.C.S. (Spl.) (Pre) 2008

I.A.S. (Pre) 1995

Ans. (a)

The optical fibre is a very thin strand of glass (silica) or plastic cable for transmitting light from one point to another. It works on the principle of total internal reflection. There is no loss of signal through an optical fibre.

32. The phenomenon used in optical fiber for transmission of light energy is

- (a) Total internal reflection
- (b) Diffraction
- (c) Scattering
- (d) Refraction

U.P.P.C.S. (Pre) 2019

Ans. (a)

See the explanation of above question.

33. Which one of the following phenomenon is used in optical fibres?

- (a) Interference
- (b) Refraction
- (c) Total internal reflection
- (d) Polarisation

U.P.P.C.S. (Mains) 2015

Ans. (c)

See the explanation of above question.

34. In fiber-optic communication, the signal travels in the form of-

- (a) Lightwave
- (b) Radio wave
- (c) Microwave
- (d) Electric wave

R.A.S./R.T.S.(Pre) 2012

Ans. (a)

Fiber-optic communication is a method of transmitting information from one place to another by sending pulses of light through an optical fibre.

35. Consider the following statements :

- 1. Fibre optics is based on the principle of total internal reflection.
- 2. In an optical fibre communication system, power consumption is extremely low.
- 3. Optical fibre communications are free from radio frequency interference.
- 4. In India, Reliance Group of Industries is engaged in the manufacture of the optical fibre.

Which of these statements are correct :

- (a) 1, 2 and 3
- (b) 1, 2 and 4
- (c) 1, 3 and 4
- (d) 2, 3 and 4

U.P. P.C.S. (Pre) 2003

U.P. U.D.A./L.D.A. (Pre) 2002

U.P. P.C.S. (Pre) 2001

Ans. (a)

Optical fibres are very thin, flexible and transparent cables made up of glass (silica) or plastic cable. They work on the principle of total internal reflection. In an optical fibre communication system, power consumption is extremely low. They are free from radio frequency and electromagnetic interferences. In India, Sterlite-Tech company started manufacturing of optical fibre in the year 2000, not Reliance Group. Thus, option (a) is correct.

36. 'Endoscope' used by the doctors for examining the inside of patient's stomach, works on the principle of-

- (a) Reflection of light
- (b) Dispersion of light
- (c) Refraction of light
- (d) Total internal reflection of light

U.P. Lower Sub. (Pre) 2009

Ans. (d)

Endoscopes are widely used to examine the interior of a hollow organ or cavity of the body. They are based on the principle of total internal reflection within a fibre optic bundle of fibres.

37. An endoscope is used by a physician to view the internal parts of a body organ. It is based on the principle of -

- (a) Refraction of light
- (b) Reflection of light

- (c) Total internal reflection of light
- (d) Dispersion of light

U.P.P.C.S. (Pre) 2010

U.P.P.C.S. (Mains) 2009

U.P.P.C.S. (Spl.) (Mains) 2008

I.A.S. (Pre) 1999

Ans. (c)

See the explanation of above question.

38. The reason of mirage is

- (a) Interference of light
- (b) Diffraction of light
- (c) Polarization of light
- (d) Total internal reflection of light

U.P.P.C.S.(Pre) 2012

U.P.P.C.S. (Pre) 1995

Ans. (d)

The mirage is caused by the refraction and total internal reflection of light at layers of air of different densities. In Desert areas, the successive upper layer is denser than those below there due to terrestrial heating. A ray of light coming from a distant object, like the top of a tree gets refracted from a denser to a rarer medium. Consequently, the refracted ray bends away from the normal until at a particular layer, the light is incident at an angle greater than the critical angle. At this stage, the incident ray suffers total internal reflection and enters the eyes of the observer. It appears as if an inverted image of the tree.

39. Mirage formation is an example of –

- (a) Refraction
- (b) Dispersion
- (c) Total internal reflection
- (d) Diffraction

U.P.P.C.S. (Mains) 2014

U.P. U.D.A./L.D.A. (Pre) 2013

U.P. Lower Sub. (Mains) 2013

Ans. (c)

See the explanation of above question.

40. Consider the following natural phenomena :

1. Terrestrial heating
2. Reflection of light
3. Refraction of light
4. Diffraction of light

Due to which of these phenomena is mirage formed ?

- (a) 1 and 2
- (b) 2, 3 and 4
- (c) 1 and 3
- (d) 4 only

I.A.S. (Pre) 2002

Ans. (c)

See the explanation of above question.

41. Which one of the following reflects back more sunlight as compared to other three?

- (a) Sand Desert
- (b) Paddy crop land
- (c) Land covered with fresh snow
- (d) Prairie land

U.P. P.C.S. (Pre) 2021

Ans. (c)

Land covered with fresh snow reflects back more sunlight as compared to other three. Fresh snow and ice reflect about 80 - 85% of the sunlight. The skin of skiers is irradiated by sunlight from all sides, since the snow does not take up the light but reflects it and remains cold. 'Albedo' (Latin meaning-whiteness) is the name given by the scientists to this ability of surfaces to reflect light. The albedo is particularly high for snow and ice.

42. The technique to integrate and mark the image of a three-dimensional object is

- (a) Audiography
- (b) Lexicography
- (c) Photography
- (d) Holography

R.A.S./R.T.S. (Pre) 1999

R.A.S./R.T.S. (Pre) 1996

Ans. (d)

Holography is a technique of producing a three-dimensional image of an object. Holography can work with sound, light or any wavelength. Holography is popularly used for preserving the transcripts and for Microfiche (a flat piece of film containing microphotographs of the pages of newspaper, catalogue or other). It is also been used for making three-dimensional feature films.

43. Which of the following is used to take 3-dimensional pictures –

- (a) Photography
- (b) Holography
- (c) Radiography
- (d) None of the above

U.P.P.C.S. (Mains) 2008

U.P.P.C.S. (Pre) 1990

Ans. (b)

See the explanation of above question.

44. The generating images on a screen by focusing an electronic beam on phosphorus coated screen is called:

- (a) Master Scan
- (b) Total Scan
- (c) Roster Scan
- (d) Radar Scan

R.A.S./R.T.S. (Pre) 1994-95

Ans. (c)

Roster scan is used in CRT (Cathode Ray Tube) monitors, in which electronic beam is projected by electronic gun to generate an image on the screen.

45. 'Raman effect' deals with the light rays passing through

- (a) Only fluids (b) Only prisms
(c) Only diamonds (d) All transparent medium

R.A.S./R.T.S. (Pre) 1992

Ans. (d)

Raman effect deals with the light rays passing through all transparent medium. According to this, a beam of light traverses a dust-free transparent sample of a chemical compound, a small fraction of light emerges in directions other than that of the incident (incoming) beam.

46. For reproducing sound, a CD (Compact Disc) audio player uses a :

- (a) Quartz crystal (b) Titanium needle
(c) Laser beam (d) Barium titanate ceramic

I.A.S. (Pre) 2000

Ans. (c)

The lens used in CD player emits ultraviolet laser beams which produce sound after reflecting through the bright surface of CD.

47. When a CD (compact disc used in audio and video systems) is seen in sunlight, rainbow-like colours is seen. This can be explained on the basis of the phenomenon of :

- (a) Reflection and diffraction
(b) Reflection and transmission
(c) Diffraction and transmission
(d) Refraction, diffraction, and transmission

I.A.S. (Pre) 2000

Ans. (a)

When a CD (Compact Disc) is viewed in sunlight, rainbow-like colours is seen. To understand this phenomenon one has to understand the construction of a CD. A CD is a thin, circular disc of metal and plastic. It is actually made of three layers. Most of a CD is made from a tough, brittle plastic called polycarbonate. Sandwiched in the middle there is a thin layer of aluminium. Finally, on the top of aluminium, is a protective layer of plastic and lacquer. It acts like a thin film, which is responsible for reflection and diffraction of light.

48. Consider the following statements :

1. Light of longer wavelength is scattered much more than the light of shorter wavelength.
2. The speed of visible light in water is 0.95 times the speed in vacuum.

3. Radio waves are produced by rapidly oscillating electrical currents.

4. To detect the overspeeding vehicles, police use the Doppler effect of reflected short radio waves.

Which of these statements are correct ?

- (a) 1 and 2 (b) 1 and 3
(c) 2 and 4 (d) 3 and 4

I.A.S. (Pre) 2002

Ans. (d)

As we know that the scattering or deviation $\delta = (n - 1) A$. Hence, the light of shorter wavelength [higher refractive index (n)] is scattered much more than the light of longer wavelength (lower ' n '). The speed of visible light in water $= 2.25 \times 10^8$ m/sec and the speed of light in vacuum $= 3.0 \times 10^8$ m/sec. Thus, it is clear that the speed of visible light in water is 0.75 times the speed of light in vacuum. So, the statement 1 and 2 are wrong while statement 3 and 4 are correct. Thus, the correct answer is option (d).

49. Assertion (A) : In a motion picture, usually 24 frames are projected every second over the whole length of the film.

Reason (R) : An image formed on the retina of eye persists for about 0.1s after the removal of the stimulus.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true but (R) is not a correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

I.A.S. (Pre) 2000

Ans. (c)

Statement (A) is correct but reason (R) is false, because the image formed on the retina of eye persists for about 1/16 of a second after the removal of the stimulus.

50. In human eye, light rays get transformed into neural impulses at :

- (a) Cornea (b) Pupil
(c) Retina (d) Lens

U.P.P.C.S. (Pre) 1997

Ans. (c)

Visual reception occurs at the retina where photoreceptor cells called cones and rods give an image colour and shadow. The image is transduced into neural impulses and then transferred through the optic nerve to the brain for processing. The brain then interprets the signals, allowing us to see.

51. The image formed on Retina is :

- (a) Equal to object but inverted
- (b) Smaller than object but straight
- (c) Smaller than object but inverted
- (d) Equal to object but straight

U.P.P.C.S. (Spl.) (Mains) 2004

U.P.P.C.S. (Pre) 1995

Ans. (c)

The image formed on Retina is real, inverted and smaller than the object. The signals for image formation are sent via the optic nerve to the brain, which ultimately processes the image and allows us to see. However, the image formed on Retina is inverted but on the basis of our experience, we see a straight image.

52. Which type of image of an object is formed by eye lens of the retina?

- (a) Erect and Real
- (b) Erect and Virtual
- (c) Inverted and Real
- (d) Inverted and Virtual
- (e) None of the above

Chhattisgarh P.C.S. (Pre) 2015

Ans. (c)

See the explanation of above question.

53. The principle reason why it is better to have two eyes than one is that:

- (a) By having two eyes we can distinguish colour easily
- (b) By having two eyes we can easily see in the dark as well in dim light.
- (c) It gives a man a type of vision known as mosaic vision.
- (d) It enhances distance and depth perception in us.

R.A.S./R.T.S. (Pre) 1994-95

Ans. (d)

Human beings have stereoscopic vision (stereopsis) means having eyes at the front of their head. The two eyes are a few centimetres apart from each other. Due to this, two eyes see the same object from two slightly different angles and send two slightly different images to the brain. The brain combines these two images to build a three-dimensional picture of the object and we can judge the depth and distance of the object more accurately.

54. Direct viewing of Sun during total solar eclipse causes irreversible damage to eyes. The retinal burn is caused by which one of the following components of the sun rays?

- (a) Heat
- (b) Rainbow light
- (c) Ultraviolet light
- (d) Infrared light

U.P.P.C.S. (Mains) 2003

Ans. (c)

The only time that the Sun can be viewed with the naked eye is during a total eclipse when the Moon completely covers the disk of the Sun. But it is never safe to look at a partial or annular eclipse or the partial phases of a total solar eclipse without proper equipment and techniques. Failure to use proper observing method may result in permanent eye damage or severe visual loss caused by the ultraviolet (UV) radiations.

55. The colour of the star is an indication of its :

- (a) Distance from the earth
- (b) Temperature
- (c) Luminosity
- (d) Distance from the sun

U.P.P.C.S. (Mains) 2008

U.P.P.C.S. (Pre) 2005

U.P.P.C.S. (Mains) 2003

Ans. (b)

The colour of the stars reflects the age and temperature of stars. Colour is reflected from the fusion reaction which indicates the temperature. If the temperature is high, then the star is young. The star will be blue during young stage which indicates high temperature. If star is red the temperature is low.

56. Diffusion of light in the atmosphere takes place due to :

- (a) Carbon dioxide
- (b) Dust particles
- (c) Helium
- (d) Water vapours

I.A.S. (Pre) 2003

U.P.P.C.S. (Pre) 2021

Ans. (b)

Sunlight is the mixture of different colours. When it passes through the atmosphere it is scattered by the air molecules, particles of dust and other subtle materials which are present in the pathway.

57. Diffusion of light in the atmosphere takes place due to

- (a) Carbon dioxide
- (b) Dust particles
- (c) Helium
- (d) None of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (b)

See the explanation of above question.

58. The colour of light is determined by its

- (a) Amplitude
- (b) Wavelength
- (c) Intensity
- (d) Velocity

U.P.P.C.S. (Mains) 2014

U.P.P.C.S. (Pre) 2012

Ans. (b)

The colour of light is determined by its wavelength. Different colours have different wavelengths. In the visible spectrum of light, the wavelength of red colour is largest and violet colour has the shortest wavelength.

59. Choose the correct statement.

- (a) Wavelength of red light is less than violet light
- (b) Wavelength of red light is more than violet light
- (c) Wavelength of violet light is more than green light
- (d) Wavelength of violet light is more than yellow light
- (e) None of the above / More than one of the above

64th B.P.C.S. (Pre) 2018

Ans. (b)

In a visible light spectrum, VIBGYOR (Violet, Indigo, Blue Green, Yellow, Orange and Red) is the order of the increasing wavelength of the various coloured lights where violet coloured light has the minimum and red coloured light has the maximum wavelength. Similarly, the same (VIBGYOR) is the order of decreasing frequency where violet light has the maximum and red light has the minimum frequency.

60. Which one of the colours has the longest wavelength?

- (a) Green
- (b) Yellow
- (c) Blue
- (d) Red

M.P.P.C.S. (Pre) 1997

Ans. (d)

See the explanation of above question.

61. The light with the shortest wavelength is

- (a) Red
- (b) Yellow
- (c) Blue
- (d) Violet

Uttarakhand P.C.S. (Pre) 2005

Ans. (d)

See the explanation of above question.

62. In the visible spectrum, the colour having the shortest wavelength is :

- (a) Red
- (b) Yellow
- (c) Blue
- (d) Violet

U.P.R.O./A.R.O. (Mains) 2014

Ans. (d)

See the explanation of above question.

63. Which one among the following has the highest energy?

- (a) Blue light
- (b) Green light
- (c) Red light
- (d) Yellow light

I.A.S. (Pre) 2009

Ans. (a)

$$\text{Energy} \propto \frac{1}{\text{wavelength}}$$

The energy of any wave is inversely proportional to its wavelength. According to VIBGYOR order, violet and after that, indigo and blue lights have the minimum wavelengths respectively, thus they have maximum energy level. So, according to the options given in the question, blue light has the maximum energy while red light has the minimum energy.

64. Which one of the following colours of white light is least deviated by the glass prism?

- (a) Green colour
- (b) Red colour
- (c) Violet colour
- (d) Orange colour

U.P. R.O./A.R.O. (Pre) (Re. Exam) 2016

Ans. (b)

Speeds of different colours of white light are same in a vacuum but differs in different matters. Thus, the refractive index of any matter differs for different colours of light. The speed of violet light is minimum and the speed of red light is maximum in the glass. So the refractive index of glass is maximum for violet light and minimum for the red light and the violet light is most deviated and red light is least deviated by the glass prism.

65. Assertion (A) : In the visible spectrum of light, red light is more energetic than green light.

Reason (R) : The wavelength of red light is more than that of green light.

Code :

- (a) Both A and R are individually true and R is the correct explanation of A
- (b) Both A and R are individually true, but R is not a correct explanation of A
- (c) A is true, but R is false
- (d) A is false, but R is true

I.A.S. (Pre) 2008

Ans. (d)

- (i) The wavelength of red light (about 700nm) is greater than the wavelength of green light (about 530 nm). Thus, reason (R) is true.
- (ii) The energy of different colors of light are inversely proportional to their wavelengths $\left(E \propto \frac{1}{\lambda}\right)$. Thus, the energy of red light (about 1.77eV) is less than the energy of green light (about 2.34 eV). So, assertion (A) is false.

66. The Sun is visible to us a few minutes before the actual sunrise because of :

- (a) Scattering of light
- (b) Diffraction of light
- (c) Total internal reflection of light
- (d) Refraction of light

R.A.S./R.T.S.(Pre) 2007

Ans. (d)

Due to the refraction of light, the Sun is visible to us a few minutes before sunrise and a few minutes after sunset.

67. The Sun and the Moon appear elliptical near the horizon because of :

- (a) Refraction
- (b) Optical illusion

- (c) Interference phenomenon
- (d) Their actual shape

U.P.P.C.S. (Spl.) (Pre) 2008

Ans. (a)

The Sun and the Moon appear elliptical near the horizon because of refraction.

68. Twinkling of stars in clear sky during nighttime can be explained with :

- (a) refraction of light (b) reflection of light
- (c) polarization of light (d) interference of light
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (a)

The twinkling of a star is due to atmospheric refraction of starlight. The starlight, on entering the Earth's atmosphere, undergoes refraction continuously before it reaches the Earth. The atmospheric refraction occurs in a medium of gradually changing refractive index. Since the atmosphere bends starlight towards the normal, the apparent position of the star is slightly different from its actual position. The star appears slightly higher (above) than its actual position when viewed near the horizon. Since the stars are very distant, they approximate point-sized sources of light. As the path of rays of light coming from the star goes on varying slightly, the apparent position of the star fluctuates and the amount of starlight entering eye flickers – the star sometimes appear brighter and sometimes fainter, which is twinkling effect.

69. Red light is used in traffic signals because :

- (a) It has the longest wavelength.
- (b) It is beautiful.
- (c) It is visible to people even with bad eyesight.
- (d) None of the above reasons.

U.P.P.C.S. (Mains) 2013

Ans. (a)

The wavelength of red colour is highest while of violet colour is lowest. Dispersal is inverse to wavelength so red light can be seen from far away more clear. That's why the red colour is used in traffic signals.

70. Assertion (A): The danger signal is made up of red colour.

Reason (R) : Red colour is scattered the least.

Select the correct answer using the codes given below:

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false.

- (d) (A) is false but (R) is true

U.P.P.C.S. (Pre) 2007

Ans. (a)

The primary reason why the red colour is used for danger signals is that red light has the highest wavelength and scattered least by air molecules so that it can be seen from far away. The appearance of the red colour of Sun at the sunset is due to the fact that neighbouring colours are scattered away in the path and light reaching the observer is predominantly red. That is why the Sun is seen red to us.

71. Red light signal is used as a danger signal because –

- (a) Red light is scattered least
- (b) This is comfortable for eyes
- (c) It produces least chemical reaction
- (d) It is least absorbed in air

U.P.P.C.S. (Mains) 2008

U.P.P.C.S. (Mains) 2011

U.P.P.C.S. (Mains) 2005

U.P.P.C.S. (Pre) 1995

Ans. (a)

See the explanation of above question.

72. The Sky appears blue due to –

- (a) Blue color is scattered most
- (b) The red color is scattered most
- (c) Blue light is minimum absorbed by atmosphere
- (d) Red light is ultimately absorbed by atmosphere

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (a)

The blue appearance of the sky is due to scattering of sunlight from the atmosphere. When we look at the sky, it is the scattered light which enters the eyes. Among the shorter wavelengths, the blue colour is present in a larger proportion in sunlight. Light of shorter wavelength is scattered by air molecules which because of their smaller size follow Rayleigh's scattering. Blue light is strongly scattered by the air molecules and reach the observer. This explains the blue colour of the sky.

73. Sky is blue because –

- (a) Blue colour in the sunlight is more than other colours
- (b) Short waves are scattered more than long waves by atmosphere
- (c) Blue colour is more absorbing to eyes
- (d) Atmosphere absorbs long wavelength more than short wavelength

Uttarakhand P.C.S. (Pre) 2005

U.P. Lower Sub. (Pre) 1998

39th B.P.S.C. (Pre) 1994

Ans. (b)

See the explanation of above question.

74. Blue colour of the sky is due to –

- (a) Scattering of light
- (b) Dispersion of light
- (c) Interference of light
- (d) Emission of more blue light as compared to other colours by the sun.

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (a)

See the explanation of above question.

75. The sky appears blue due to –

- (a) Diffraction of light
- (b) Reflection of light
- (c) Refraction of light
- (d) Scattering of light

U.P.P.C.S. (Mains) 2011

Ans. (d)

See the explanation of above question.

76. Consider the following statements :

1. The Sky appears blue because of scattering.
2. The main cause of seven colors of the rainbow is scattering of light.
3. At the time of sunrise, the sun appears red, due to scattering of light.

Choose the correct statement/s from the given options:

- (a) Only 1 is true
- (b) 1 and 3 is true
- (c) Only 2 is true
- (d) 2 and 3 is true

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (b)

The intensity of scattered light is inversely proportional to its wavelength and the wavelength of violet and blue is low, thus they scattered more. This is the main cause for the blue colour of the sky. As we know that sunlight consists of seven colours, of these, red has the maximum wavelength. During sunrise and sunset, the rays have to travel a larger part of the atmosphere because they are very close to the horizon. Therefore light other than red is mostly scattered away. Most of the red light which is least scattered enters our eyes. Hence, the sun appears red. The rainbow is caused by reflection, refraction and dispersion of light in water droplets resulting in a spectrum of light appearing in the sky. Thus, statements 1 and 3 are correct.

77. The red colour of the Sun at sunset and sunrise is due to –

- (a) Scattering of light
- (b) Refraction of light
- (c) Total internal reflection of light
- (d) Dispersion of light

U.P.P.C.S. (Mains) 2013

U.P. P.C.S. (Pre) 1992

Ans. (a)

See the explanation of above question.

78. Read the following statements and choose the correct option :

(i) For passengers flying at very high altitudes, sky appears dark.

(ii) Scattering is not prominent at such heights.

- (a) (i) and (ii) both are true, and (ii) is correct reasoning of (i)
- (b) (i) and (ii) both are true, but (ii) is not correct reasoning of (i)
- (c) (i) is correct and (ii) is incorrect
- (d) (i) is incorrect and (ii) is correct

Chhattisgarh P.C.S. (Pre) 2022

Ans. (a)

The scattering of the light takes place because of the particles present in the atmosphere. At a higher altitude, the atmospheric medium is very rarer and thus, scattering of light is very low or not prominent. Hence, the sky appears dark to the passengers flying at high altitudes. So, statements (i) and (ii) both are true, and (ii) is correct reasoning of (i).

79. An astronaut sees the colour of the sky as –

- (a) Blue
- (b) White
- (c) Black
- (d) Red

U.P.P.C.S. (Mains) 2005

Ans. (c)

The colour of the sky for an astronaut is black. It is because of the fact that in the space there is no atmosphere. In space, there are no air particles against which the sunlight can strike, get reflected or scattered. Hence, astronaut see everything black.

80. When we see the sky from a plane in high altitude, then it seems –

- (a) Blue
- (b) Black
- (c) White
- (d) Red

U.P.P.S.C. (R.I.) 2014

Ans. (b)

See the explanation of above question.

81. Sea seems blue due to –

- (a) Excess deepness
- (b) Reflection of sky and scattering of light by the drops of water
- (c) Blue colour of water
- (d) Upper layer of water

40th B.P.S.C. (Pre) 1995

Ans. (b)

The sea looks blue due to reflection of sky and scattering of light by the drops of water. Light scattered as it passes through air or water. Since blue light has a short wavelength, it scatters most easily.

82. The (Green) Grass appears green to us because :

- (a) It reflects green colour light to our eyes
- (b) It absorbs green colour light
- (c) It reflects all other colour lights except that of green colour
- (d) None of the above

U.P. P.C.S. (Mains) 2017

Ans. (a)

The colour we see on the surface of an object is not actually the colour in that surface. If a surface looks green, it means that it absorbs all colours of the spectrum except green. It looks black to our eyes in case no colour is reflected. If an object is white it reflects all light.

83. When a soap film on the water is seen in the daytime, it shows beautiful colours. This phenomenon is due to:

- (a) diffraction of light
- (b) refraction of light
- (c) polarization of light
- (d) interference of light
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (d)

The colours that we see when observing a soap film are caused by interference of light reflecting off of two nearby surfaces – the outer surface and the inner surface. Interference patterns are created when two reflected waves line up in phase or out of phase. White light is a mixture of all the wavelengths of the visible spectrum and the interference enhances some wavelengths and diminishes others, creating visible colours. The specific wavelengths that are enhanced or diminished are determined by the thickness of the film and its index of refraction.

84. Suppose a rocketship is proceeding from the Earth at a speed of 2/10th the velocity of light. A light in the rocketship appears blue to the passengers on the ship. What colour would it appear to an observer on the Earth?

- (a) Blue
- (b) Orange
- (c) Yellow
- (d) Yellow-Orange

I.A.S. (Pre) 1995

Ans. (d)

To determine the color that would appear to an observer on the Earth, we use the following formula.

$$v = \pm (\lambda - \lambda_0 / \lambda) C$$

Where λ = Prime wavelength

λ_0 = Virtual wavelength

v = Velocity of rocket

C = Velocity of light

Because the rocket is proceeding from the Earth, thus positive sign will be taken.

$$\frac{2}{10} C = \left[\frac{\lambda - \lambda_0}{\lambda} \right] C$$

$$\Rightarrow \frac{2}{10} = \left[\frac{\lambda - \lambda_0}{\lambda} \right]$$

$$\Rightarrow \frac{2}{10} = 1 - \frac{\lambda_0}{\lambda}$$

$$\Rightarrow \frac{\lambda_0}{\lambda} = 1 - \frac{2}{10}$$

$$\Rightarrow \frac{\lambda_0}{\lambda} = \frac{8}{10}$$

$$\Rightarrow \lambda = \frac{10}{8} \lambda_0 = 1.25 \lambda_0$$

Because the wavelength of blue colour lies between 450nm to 490 nm. Thus, the wavelength of the light would appear to an observer on the Earth has a range between 562.5 nm to 612.5 nm, which will be yellow-orange light.

85. What will be the nature of image formed by a convex mirror when the position of object is between infinity and pole P of the mirror?

- (a) Virtual and erect
- (b) Real and erect
- (c) Real and inverted
- (d) Virtual and inverted

Chhattisgarh P.C.S. (Pre) 2020

Ans. (a)

When the position of object is between infinity and pole (P) of the convex mirror, then the image will be formed between the pole (P) and the focus (F) behind the mirror. The image thus formed is diminished, virtual and erect.

86. Large number of thin strips of black paint are made on the surface of a convex lens of focal length 20 cm to catch the image of a white horse. The image will be:

- (a) a horse of less brightness
- (b) a zebra of black stripes
- (c) a horse of black stripes
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

When light passes through a convex lens, it converges or comes together at a focal point. In this case, the convex lens has a focal length of 20 cm. When the lens has a large number of thin black strips on its surface, these strips will block or absorb some of the incident light. As a result, less light will reach the focal point and form the image. Hence, thin black strips on the surface of a convex lens will only reduce the intensity of the light but will not cast any shadow on the image, as it is quite thin. Thus, option (a) is the correct answer.

87. In driving a car which type of mirror would you like to use for observing traffic at your back ?

- (a) Concave mirror (b) Spherical mirror
(c) Convex mirror (d) Plain mirror

Uttarakhand P.C.S. (Pre) 2005

U.P. P.C.S. (Pre) 1991

Ans. (c)

A convex mirror gives a much wider field of view of the traffic at your back than a plain mirror of the same size. It always forms an erect and diminished image and they also have a wider field of view as they are curved outwards.

88. Which one of the following types of the mirror is used to see the traffic behind the car?

- (a) Convex mirror (b) Concave mirror
(c) Plane mirror (d) Spherical mirror

U.P.P.C.S. (Pre) 2015

Ans. (a)

See the explanation of above question.

89. Which mirror is used as a rear view mirror in vehicles ?

- (a) Plain (b) Convex
(c) Concave (d) Cylindrical

U.P.P.C.S (Pre) 2011

U.P.P.C.S. (Mains) 2010

Ans. (b)

See the explanation of above question.

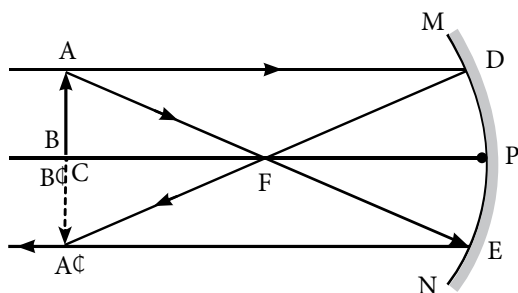
90. The image formed by concave mirror is real, inverted and of the same size as that of the object. The position of the object should be :

- (a) at the focus
(b) at the centre of curvature
(c) between the focus and centre of curvature
(d) beyond the centre of curvature

69th B.P.S.C. (Pre) 2023

Ans. (b)

If an object is placed at the centre of curvature of a concave mirror, then the image formed is real, inverted and of the same size as that of the object. The image formed is not magnified.



91. A person standing in front of a mirror finds his image larger than himself. This implies that the mirror is :

- (a) concave
(b) plane
(c) convex
(d) cylindrical with bulging side outwards
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (a)

A person standing in front of a mirror finds his image larger than himself implies that the mirror is concave. A spherical mirror, whose reflecting surface is curved inwards, that is, faces towards the centre of the sphere, is called a concave mirror. Concave mirrors are commonly used in torches, search-lights and vehicles' headlights to get powerful parallel beams of light. They are often used as shaving mirrors to see a large image of the face. Dentists use concave mirrors to see large images of the teeth of patients.

92. For shaving, one uses –

- (a) Concave mirror (b) Plain mirror
(c) Convex mirror (d) None of these

43rd B.P.S.C. (Pre) 1999

Ans. (a)

People use a concave mirror for shaving because when a man standing between the principal focus and pole of a concave mirror, he sees as enlarged, the erect and virtual image of his face. This is the reason why a concave mirror of a large focal length is used for shaving.

93. The mirror, which is used in searchlights, is –

- (a) Concave mirror (b) Convex mirror
(c) Simple mirror (d) None of these

U.P.P.C.S.(Mains) 2009

Ans. (a)

Concave mirror is used in the headlights of cars and in searchlights. A light placed at the focus of a concave lens can form a parallel beam and is used in car headlights and searchlights.

94. A dentist mirror is a –

- (a) Cylindrical mirror (b) Concave mirror
(c) Convex mirror (d) Plane mirror

U.P.P.C.S. (Mains) 2009

Ans. (b)

Concave mirrors are used by dentists because at a short range (object distances less than the focal length) they produce magnified, upright images. It is useful to have a magnified image of a tooth when you're looking for or repairing cavities, cracks or other abnormalities.

95. Which of the following is used by the dentist to see the image of teeth of a patient?

- (a) Concave lens (b) Convex lens
(c) Convex mirror (d) Plane mirror
(e) None of these

Chhattisgarh P.C.S. (Pre) 2015

Ans. (e)

See the explanation of above question.

96. The mirror used by a dentist to examine the teeth of patients is

- (a) Concave (b) Convex
(c) Plane (d) Cylindrical

U.P.P.C.S.(Pre) 2013

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (a)

See the explanation of above question.

97. In order to avoid sunlight, which of the following colour combination for an umbrella is most appropriate?

- (a) Top black & bottom white
(b) Top white & bottom black
(c) Only black
(d) Only white

U.P.P.C.S. (Pre) 1993

Ans. (b)

In order to avoid sunlight/heat, option (b) top white & bottom black combination for an umbrella is most appropriate. The outer colour (white) will reflect most of the radiations, while the black colour inside will absorb anything passing through the outer layer.

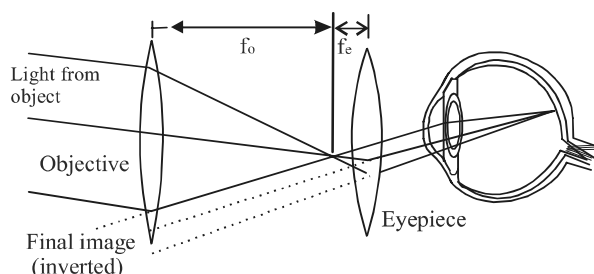
98. The image formed by astronomical telescope is

- (a) Virtual and diminished
(b) Virtual and magnified
(c) Real and diminished
(d) Real and magnified

U.P.P.C.S. (Mains) 2013

Ans. (b)

Astronomical telescope consists of two convex lenses: Objective and Eyepiece.



The final image formed by the astronomical telescope is inverted, virtual and highly magnified.

99. As the diameter of objective lens of a telescope increases, the resolution of telescope :

- (a) Remains same
(b) Depends on focal length of lens
(c) Increases
(d) Decreases

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (c)

Resolution or resolving power is the ability of a telescope to distinguish between two closely spaced objects. The objective lens is the primary lens in a refracting telescope that gathers light from a distant object. A larger objective lens gathers more light, which brightens the image and improves the contrast, making it easier to resolve details. With a larger objective lens, the telescope can see smaller details and finer features in the image. The resolution of a telescope is limited by diffraction, a phenomenon where light waves bend around obstacles. A larger aperture (diameter) helps reduce the effects of diffraction, leading to sharper images. Thus, as the diameter of the objective lens of a telescope increases, its resolving power also increases, allowing it to distinguish finer details.

100. The number of images of an object placed between two parallel mirrors is :

- (a) Two (b) One
(c) Six (d) Infinite

U.P.P.C.S.(Mains) 2013

U.P.P.C.S. (Pre) 1994

Ans. (d)

When an object is placed in between two parallel plane mirrors, then the number of images formed will be infinite. Each image will be the result of an image. When two plane mirrors are inclined at an angle of θ , then the number of images formed by the object between them is $\left(\frac{360^\circ}{\theta} - 1\right)$ (If the value of $\frac{360^\circ}{\theta}$ is even), or $\left(\frac{360^\circ}{\theta}\right)$ (If the value of $\frac{360^\circ}{\theta}$ is odd).

101. How many images will be formed if a point light source is placed between two parallel plane mirrors?

- (a) Two (b) Four
(c) Eight (d) Infinite

U.P.P.C.S. (Pre) 2015

Ans. (d)

See the explanation of above question.

102. When a mirror is rotated by an angle θ , the reflected ray will rotate by :

- (a) 0 (b) $\theta/2$

(c) θ

(d) 2θ

I.A.S. (Pre) 1996

Ans. (d)

If a plane mirror is rotated through a certain angle, then the reflected ray rotates through twice of that angle. So if a mirror is rotated by an angle θ , the reflected ray will rotate by 2θ .

103. Assertion (A): Small glass beads fixed on traffic signals glow brightly when light falls upon them.

Reason (R) : Light is totally reflected when the angle of incidence exceeds a certain critical value and light travelling in a denser medium is reflected from a rarer medium.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (b) Both (A) and (R) are true but (R) is not a correct explanation of (A)
- (c) (A) is true but (R) is false
- (d) (A) is false but (R) is true

I.A.S. (Pre) 2000

Ans. (a)

When light travelling through a denser medium to rarer medium, then the light is totally reflected when the angle of incidence exceeds a certain critical value, which is called total internal reflection. Here, small glass beads are denser medium, so due to the phenomenon of total internal reflection, traffic signals glow brightly when light falls upon them. Thus, both (A) and (R) are true and (R) is the correct explanation of (A).

104. Light is made of seven colours. What is the method of separating the colours?

- (a) The colours can be separated by a prism
- (b) The colours can be separated by a filter
- (c) The colours can be separated by plants
- (d) The colours cannot be separated

47th B.P.S.C. (Pre) 2005

Ans. (a)

Colours could be separated by using a prism. The white colour is made up of all seven colours. When it passes through a prism, due to different wavelengths, different colours of light is refracted by a different amount.

105. The correct order of colours in a rainbow is :

- (a) Blue, green, violet
- (b) Violet, green, blue
- (c) Blue, yellow, green
- (d) Blue, green, yellow

R.A.S./R.T.S.(Pre) 1999

Ans. (d)

A rainbow is a meteorological phenomenon that is caused by reflection, refraction, and dispersion of light in water droplets resulting in a spectrum of light appearing in the sky. The correct order of colours in a rainbow is Violet, Indigo, Blue, Green, Yellow, Orange and Red (VIBGYOR).

106. Rainbow is produced when sunlight falls on drops of rain. which of the following physical phenomena are responsible for this?

1. Dispersion

2. Refraction

3. Internal reflection

Select the correct answer using the code given below :

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

I.A.S. (Pre) 2013

Ans. (d)

A rainbow is an optical phenomenon caused by refraction, internal reflection and dispersion of light in water droplets resulting in a continuous spectrum of light appearing in the sky in a direction directly opposite the Sun. The water droplets act like small prisms. They refract and disperse the incident sunlight causing the white light to split in different colours and then reflect it internally and finally refract it again at multiple angles when it comes out of the raindrop.

107. Dispersion of light is possible by :

- (a) Prism
- (b) Convex lens
- (c) Concave lens
- (d) Simple mirror
- (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

Dispersion of light is possible by prism. Upon passage through the prism, the white light is separated into its different component colours, which is known as dispersion.

108. Which one of the following colours is seen in the middle of a Rainbow ?

- (a) Blue
- (b) Green
- (c) Red
- (d) Yellow

U.P.P.C.S. (Mains) 2012

Ans. (b)

The correct order of colors of the rainbow is Red, Orange, Yellow, Green, Blue, Indigo and Violet. Thus it is clear that 'Green' Color is seen in the middle of a rainbow (VIBGYOR).

109. The terminal colour bands of the rainbow are :

- (a) Blue and Orange
- (b) Green and Red
- (c) Yellow and Green
- (d) Violet and Red

U.P. Lower Sub. (Mains) 2015

Ans. (d)

A rainbow is a meteorological phenomenon that is caused by reflection, refraction, and dispersion of light in water droplets resulting in a spectrum of light appearing in the sky. The apparent discreteness of main colours is Red, Orange, Yellow, Green, Blue, Indigo and Violet. Hence, option (d) is the correct answer.

110. In which direction the rainbow is seen at 12 noon?

- (a) In the West (b) In the South
(c) In the East (d) It cannot be seen

43rd B.P.S.C. (Pre) 1999

Ans. (d)

A rainbow is located opposite to the sun; this explains why rainbow is not seen at 12 noon with the sun overhead.

111. Which of the following is responsible for production of shadow ?

- (a) Polarisation
(b) Rectilinear propagation of light
(c) Diffraction
(d) Interference

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (b)

Shadows are formed when an opaque object obstructs the path of light, preventing it from reaching a surface or area behind the object. Light travels in straight lines, and when it encounters an opaque object, it cannot pass through, creating a dark region behind the object. The shape of the shadow is determined by the shape of the object blocking the light and the position of the light source relative to the object. Hence, rectilinear propagation of light is responsible for forming a shadow.

112. The reason for a swimming pool to appear less deep than the actual depth is :

- (a) Reflection (b) Light scattering
(c) Refraction (d) None of the above

U.P. R.O./A.R.O. (Pre) 2023

Ans. (c)

A swimming pool appears less deep than its actual depth because light rays from the bottom of the pool are bent away from the normal as they pass from water into the air (denser to rarer medium). This phenomenon is called the refraction of light. Due to this the image of the bottom of the pool seems nearer to us and the pool appears less deep.

113. Assertion (A) : A stick is dipped in water in a slanting position. If observed sideways, the stick appears short and bent at the surface of the water.

Reason (R) : A light coming from the stick undergoes scattering from water molecules giving the stick a short and bent appearance.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true but (R) is not a correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

I.A.S. (Pre) 2001

Ans. (c)

When a stick is dipped in water in a slanting position, it appears short and bent at the surface of water. It is caused by the phenomenon of refraction of light. When you immerse a stick in water, you are actually putting and watching it from rarer medium to denser medium. When the rays of light pass from denser to rarer medium they refract and move away from the normal and reach the person's eye. Hence, stick appears to be bent and short when immersed in water due to refraction of light. Hence, (A) is true but (R) is false.

114. Consider the following statements :

- 1. If a person looks at a coin which is in a bucket of water, the coin will appear to be closer than it really is.**
- 2. If a person under water looks at a coin above the water surface, the coin will appear to be at a higher level than it really is.**

Which of the above statements is/are correct ?

- (a) 1 and 2 (b) 1 only
(c) 2 only (d) Neither 1 nor 2

I.A.S. (Pre) 1999

Ans. (a)

Statement 1 is true. The coin will appear to be closer than it really is due to the refraction of light. In this case, the ray that is coming to the eyes is traveling from denser medium (water) to the rarer medium (air). The statement 2 is also true. If a person underwater looks at a coin above the water surface the coin will appear to be at a higher level than it really is also due to the refraction of light. In this case, the ray is travelling from the rarer medium to denser medium.

115. An air bubble in water will act like a :

- (a) Convex mirror (b) Convex lens
(c) Concave mirror (d) Concave lens

U.P. Lower Sub. (Pre) 2013

U.P. U.D.A./L.D.A. (Spl.) (Pre) 2010

U.P.P.C.S. (Spl.) (Pre) 2008

I.A.S. (Pre) 1995

Ans. (d)

An air bubble in water, whose surface is convex will act like a concave lens (divergent lens). The refractive index of water is more than that of air. Thus the nature of lens of submerged water bubble is changed.

116. An air bubble inside water behaves as a:

- (a) Bifocal lens (b) Convergent lens
(c) Divergent lens (d) Cylindrical lens

U.P.P.C.S. (Pre) 2002

Ans. (c)

See the explanation of above question.

117. Two glass lenses of same focal length, one being convex and the other concave, are kept in contact, the combination will act as :

- (a) convergent lens (b) divergent lens
(c) plane glass sheet (d) mirror

U.P.P.S.C. (GIC) 2017

Ans. (c)

If two glass lenses of same focal length, one being convex and the other concave are kept in contact then resultant focal length of the combination will be infinity and it will act as a plane glass sheet.

118. The minimum height of a plane mirror to see the full size image of a person is equal to –

- (a) The height of the person
(b) Half the height of the person
(c) One-fourth the height of the person
(d) Double the height of the person

U.P.P.C.S. (Pre) 2014

U.P.P.C.S. (Pre) 2011

U.P.P.C.S. (Mains) 2011

U.P. U.D.A./L.D.A. (Spl.) (Mains) 2010

Ans. (b)

The minimum height of a plane mirror to enable a person to see full image is 1/2 of his height. According to the principle of reflection, the angle of incidence is equal to angle of reflection. When the ray of light travels from bottom to top of a person and reaches his eyes then he is able to see the clear image. The ray from the foot strikes the mirror at its bottom and reaches his eye after reflecting from the mirror. Thus, the height of the mirror must equal to half the height of the person.

119. The minimum height of mirror required to see whole stand up image of a man of height 1.50 metre shall be

- (a) 0.75 metre (b) 1.00 metre
(c) 1.50 metre (d) 3.00 metre

U.P.P.S.C. (R.I.) 2014

Ans. (a)

In order to see full-length image, the minimum height of the mirror required is half of a person's height.

Therefore, minimum height of mirror required = $1.50/2$ metre
= 0.75 metre.

120. The radius of curvature of the plane mirror is :

- (a) Zero (b) One
(c) Infinity (d) Between one and infinity

U.P. Lower Sub. (Pre) 2015

Ans. (c)

A plane mirror is a spherical mirror with an infinite radius of curvature. As we know that the reflection in the plane mirror forms at the same distance at which the object is placed in front of the mirror. That is,

d_o = distance of the object from the mirror

d_i = distance of reflection from the mirror

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{2}{r} \quad (\text{here } r = \text{radius of curvature})$$

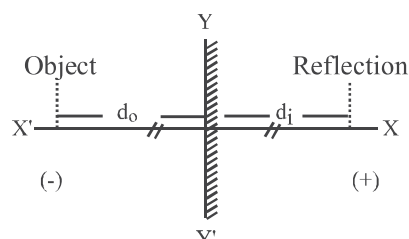
But for the plane mirror

$$d_o = -d_i$$

$$\text{thus } \frac{2}{r} = 0$$

$$\text{or, } r = \frac{2}{0}$$

$$\text{or } r = \infty$$



Thus, the radius of curvature of plane mirror is considered as infinity.

121. What type of lens is used in magnifying glass?

- (a) Concave lens (b) Plano-concave lens
(c) Convex lens (d) Convex mirror
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (c)

Convex lenses are used in magnifying devices due to their magnifying properties. A convex lens has the capacity to make an object appear much larger than it actually is. Hence, it is used in the magnifying glass. Magnifying glass is used to enlarge items such as writing so it can be read more easily.

122. To remove the defect of long sightedness one uses :

- (a) Concave lens (b) Convex mirror

- (c) Convex lens (d) Concave mirror

43rd B.P.S.C. (Pre) 1999
Chhattishgarh P.C.S. (Pre) 2016

Ans. (c)

Far-sightedness (hypermetropia) as it is medically termed is a vision condition in which distant objects are usually seen clearly, but near objects are not clearly visible. To remove this vision problem one should use a convex lens.

123. Which one of the following statements is **not** true for a person suffering from hypermetropia?

- (a) The person can see far objects distinctly.
(b) The focal length of the lens is large.
(c) The image of the close objects is focused behind the retina.
(d) A concave lens is used to correct this defect.

U.P.P.C.S. (Pre) 2010

Ans. (d)

Far-sightedness (hypermetropia) is a common vision problem, affecting about a fourth of the population. People with hypermetropia can see distant objects very well but have difficulty in focusing on objects that are close. This vision problem occurs when light rays entering the eye are focused behind the retina rather than directly on it. The eyeball of a farsighted person is shorter than normal. A convex lens is used to correct this defect.

124. Reading glasses are made from which type of lenses?

- (a) Concave (b) Convex
(c) Plain (d) None of these

44th B.P.S.C. (Pre) 2000

Ans. (b)

A convex lens is thicker in the middle and thinner at the edges. Rays of light that pass through the lens are brought closer together. A convex lens is also called a converging lens. A convex lens is used in reading glasses and it is also used to remove the defect of far-sightedness.

125. Which of the following lenses is used to minimize Myopia?

- (a) Convex lens (b) Concave lens
(c) Cylindrical lens (d) None of these

U.P.P.C.S. (Mains) 2011

M.P.P.C.S. (Pre) 2008

Ans. (b)

A concave lens is used to minimize the defect of myopia.

126. A myopic eye can be corrected by using a :

- (a) Convex lens (b) Concave lens
(c) Cylindrical lens (d) Plane Convex lens

U.P.P.C.S. (Mains) 2008

U.P. Lower Sub. (Pre) 2002

Ans. (b)

Myopia or near-sightedness is a refractive error which means that the eye does not bend or refract light properly to a single focus to see the image clearly. In Myopia, close objects look clear but distant objects appear blurred. It is corrected by using a concave lens.

127. Myopia is corrected by using :

- (a) Concave lens (b) Concave mirror
(c) Convex lens (d) Convex mirror
(e) None of these

Chhattishgarh P.C.S. (Pre) 2016

Ans. (a)

See the explanation of above question.

128. For the myopic eye, the defect is cured by :

- (a) Convex lens (b) Concave lens
(c) Cylindrical lens (d) None of the above

U.P.P.C.S. (Mains) 2013

Ans. (b)

See the explanation of above question.

129. Myopia and Hypermetropia can be respectively corrected by :

- (a) Plano-concave and Cylindrical lens
(b) Convex and Concave lens
(c) Concave and Convex lens
(d) Concave and Cylindrical lens

Rajasthan P.C.S. (Pre) 2023

Ans. (c)

Myopia (near-sightedness) and Hypermetropia (far-sightedness) can be respectively corrected by concave and convex lenses.

130. When a person enters into a dark room from the strong light area, he is not able to see clearly for some time. Later he gradually begins to see things. This is because-

- (a) Changes in the size of pupil
(b) Changes in the diameter and focal length of lens
(c) Bleaching and reformation of Rhodopsins
(d) Eyes become familiar with darkness in course of time

R.A.S./R.T.S. (Pre) 2008

Ans. (*)

The amount of light entering the eye through the pupil is controlled by the iris. But this adjustment takes some time; for example, our eyes cannot see everything clearly when we enter a dark room from a bright light because in bright light, the size of the pupil of our eye is small. When we enter the dark room, due to the small size of the pupil, very little light enters our eyes. But after some time, the pupil expands which permits more light to enter the eyes and we become able to see things.

So the option (a) is correct. Also it takes some time for the eyes to adapt with darkness and the reason behind this are rods & cones. These are two types of cells in our retina that absorb light. The rods help us to see in the dark while cones help us to see under bright conditions. So, when a person moves from a well lit area to a dark from a well lit area to a dark area, initially he's not able to see clearly because cones cease functioning in low light. Also, the rod pigments (rhodopsins) have been bleached out due to the bright light and these cells cannot function normally again until the bleach pigments are restored back to their unbleached state and this takes time. Hence, option (c) and (d) are also correct for this question.

131. The focusing in the eye is done by :

- (a) Change in the convexity of the lens
- (b) To and fro movement of the lens
- (c) To and fro movement of the retina
- (d) Change in the refractive index of the eye fluid

U.P.P.S.C. (R.I.) 2014

Ans. (a)

We see the world around us because of the way our eyes bend (refract) light. The refractive or bending power of the cornea and humor is constant. However, that of the lens can be changed by changing its shape. That is by making it more or less convex so that light can be properly focused on the retina. The greater the lens convexity or bulge the more it bends the light. The flatter the lens, the less it bends that light.

132. The change of focal length of an eye lens is caused by action of the :

- (a) Pupil
- (b) Retina
- (c) Ciliary muscles
- (d) Iris
- (e) None of the above

Chhattisgarh P.C.S. (Pre) 2014

Ans. (c)

The eye lens is composed of a fibrous, jelly-like material. Its curvature can be modified to some extent by the ciliary muscles. The change in the curvature of the eye lens can thus change its focal length. When the muscles are relaxed, the lens becomes thin. Thus its focal length increases. This enables us to see distant objects clearly. When you are looking at objects closer to the eye, the ciliary muscles contract. This increases the curvature of the eye lens. The eye lens then becomes thicker. Consequently, the focal length of the eye lens decreases. This enables us to see nearby objects clearly.

133. Which one of the following statement is not correct ?

- (a) Mirage is an optical illusion arising due to total internal reflection of light
- (b) Concave mirrors are used in searchlights
- (c) In summer, white or light coloured clothes are preferred to dark coloured clothes generally

- (d) Convex mirror with a small aperture is used by doctors to examine, ear, nose or throat of a patient.

U.P.P.C.S. (Mains) 2010

Ans. (d)

Head Mirror is mostly used by doctors for the examination of ear, nose, and throat. It comprises a circular concave mirror, with a small hole in the middle and is attached to a head-band. Hence statement (d) is not correct. Remaining statements are correct.

134. The least distance of distinct vision is (cm.) :

- (a) 25
- (b) 5
- (c) 75
- (d) 100

U.P.P.C.S. (Mains) 2008

Ans. (a)

The normal human eye can focus a sharp image of an object on the eye if the object located anywhere from infinity to a certain point called the "Near point". The minimum distance of an object from eye to have its clear image is called "least distance of distinct vision". It is about 25 cm for the normal eye and denoted by 'D'.

135. For normal eyes what is the minimum distance for clear vision ?

- (a) 10 cm.
- (b) 15 cm.
- (c) 20 cm.
- (d) 25 cm.
- (e) 30 cm.

Chhattisgarh P.C.S. (Pre) 2013

Ans. (d)

See the explanation of above question.

136. The minimum distance of clear vision for a normal healthy eye of human being is supposed to be :

- (a) 50 cm
- (b) 10 cm
- (c) 15 cm
- (d) 25 cm

M.P. P.C.S. (Pre) 2016

Ans. (d)

See the explanation of above question.

137. The power of a lens is measured in

- (a) dioptre
- (b) aeon
- (c) lumen
- (d) candela

M.P.P.C.S. (Pre) 2017

Ans. (a)

Dioptre is the unit of measurement for the refractive power of a lens. The power of a lens is defined as the reciprocal of its focal length in metres.

$$D = \frac{1}{f}$$

where D is power of lens in dioptre and f is the focal length in metres. Converging lens has positive optical power and diverging lens has negative optical power.

138. The power of lens is measured in :

- (a) Watt (b) Ampere
(c) Volt (d) Dioptré
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (d)

See the explanation of above question.

139. If the lens power of glass is +2 dioptré, its focal length will be –

- (a) 200 cm (b) 100 cm
(c) 50 cm (d) 2 cm

41st B.P.S.C. (Pre) 1996

Ans. (c)

We know that,

$$D = \frac{1}{f}$$

$$1 \text{ metre} = 100 \text{ cm}$$

$$\text{or } f = \frac{100}{D} \text{ cm}$$

$$f = \frac{100}{2} = 50 \text{ cm}$$

140. Power of the sunglass is –

- (a) 0 Dioptré (b) 1 Dioptré
(c) 2 Dioptré (d) 4 Dioptré

U.P.P.C.S. (Pre) 1995

Ans. (a)

The power of the sunglass is 0 dioptré.

141. Two slits are made one millimetre apart and the screen is placed one metre away. What is the fringe separation when blue-green light of wavelength 500 nm is used?

- (a) 0.5 m (b) 0.5 mm
(c) 50 mm (d) 50 m

Uttarakhand P.C.S. (Pre) 2024

Ans. (b)

In the context of Young's double-slit experiment, fringe separation (β), also known as fringe width, is the distance between two consecutive bright or dark fringes in an interference pattern, calculated as $\lambda D/d$, where D is the distance to the screen, and d is the slit separation.

Given that, $\lambda = 500 \text{ nm} = 5 \times 10^{-7} \text{ m}$

$$D = 1 \text{ m}$$

$$d = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$$

$$\begin{aligned} \text{Hence, fringe separation } (\beta) &= \frac{\lambda D}{d} \\ &= \frac{5 \times 10^{-7} \times 1}{1 \times 10^{-3}} \text{ m} \\ &= 5 \times 10^{-4} \text{ m} \\ &= 0.5 \times 10^{-3} \text{ m} = 0.5 \text{ mm} \end{aligned}$$

142. Which one of the following combinations of aperture and shutter speed of a camera will allow the maximum exposure?

- (a) $f=22$, $1/60$ (b) $f=16$, $1/125$
(c) $f=8$, $1/250$ (d) $f=5.6$, $1/1000$

I.A.S. (Pre) 1994

Ans. (c)

The exposure value EV is

$$EV \propto N^2 \times t$$

where N is the relative aperture $= \frac{1}{f}$,
and, t is the exposure time

$$\text{For, } f=22, \frac{1}{60}$$

$$EV \propto \left(\frac{1}{22}\right)^2 \times \left(\frac{1}{60}\right)$$

$$\therefore EV \approx 0.0021 \times 0.0167 \approx 0.000035$$

$$\text{For, } f=16, \frac{1}{125}$$

$$EV \propto \left(\frac{1}{16}\right)^2 \times \frac{1}{125}$$

$$\therefore EV \approx 0.0039 \times 0.008 \approx 0.000031$$

$$\text{For, } f=8, \frac{1}{250}$$

$$EV \propto \left(\frac{1}{8}\right)^2 \times \frac{1}{250}$$

$$\therefore EV \approx 0.0156 \times 0.004 \approx 0.000062$$

$$\text{For, } f=5.6, \frac{1}{1000}$$

$$EV \propto \left(\frac{1}{5.6}\right)^2 \times \frac{1}{1000}$$

$$\therefore EV \approx 0.0318 \times 0.001 \approx 0.000032$$

Hence, among the given options, (c) has maximum exposure value.

143. Retina of the eye is comparable to which of the following parts of a traditional camera?

- (a) Film (b) Lens
(c) Shutter (d) Cover

U.P.P.C.S. (Pre) 2011

Ans. (a)

The back of your eye is lined with a layer known as the retina. The retina operates in a similar way to the film inside a traditional (non-digital) camera. The middle part of the retina (known as the macula) takes care of your sharp vision. Your retina's nerve cells change light rays into electrical impulses and then sends them to the brain through the optic nerve.

144. In human eye, image of object forms at :

- (a) Cornea (b) Iris
(c) Pupil (d) Retina

M.P.P.C.S. (Pre) 2017, 2014

Uttarakhand Lower Sub. (Pre) 2010

Ans. (d)

The functioning of the human eye is similar to that of the photographic camera. The light rays travel into the cornea and are refracted by the cornea and lens. In a normal eye, the image formed are transmitted from the retina to the brain along the optic nerve, where data processing takes place (the image is perceived the right way up).

145. The human eye forms the image of an object at its –

- (a) Cornea
- (b) Iris
- (c) Pupil
- (d) Retina
- (e) None of the above

Chhattisgarh P.C.S. (Pre) 2014

Ans. (d)

See the explanation of above question.

146. In long-sightedness, images are formed at following :

- (a) On Retina
- (b) In front of Retina
- (c) Behind Retina
- (d) On Blind Spot

U.P. R.O./A.R.O. (Mains) 2017

Ans. (c)

In long-sightedness (Hypermetropia) the image of a nearby object is formed behind the retina. It results in close objects appearing blurry. The condition occurs in case the eyeball is too short or the focal length of the eye lens has been increased. The long-sightedness is corrected by the convex lens.

147. The colour of an opaque object is due to the colour it–

- (a) Absorbs
- (b) Refracts
- (c) Reflects
- (d) Scatters

I.A.S. (Pre) 1994

Ans. (c)

The colour of an opaque object is the same as the colour of the light that is reflected by it. Opaque object doesn't pass the light through it. It absorbs some part of incident light and reflects the other part. The reflected part of light decides the colour of that object.

148. Colour in a colour television is produced by a combination of three basic colours, they are :

- (a) Red, blue and orange
- (b) Red, green and blue
- (c) Red, yellow and green
- (d) Red, green and brown

R.A.S./R.T.S. (Pre) 1992

Ans. (b)

The cathode ray tube (CRT), which is used in colour television uses three different phosphors which emit red, green and blue light respectively. They are packed together in stripes or clusters called "triads".

149. Which primary colours produce different shades on TV screen ?

- (a) Red, green and blue
- (b) Red, yellow and green
- (c) Orange, green and purple
- (d) Red, yellow, green and black

Uttarakhand Lower Sub. (Pre) 2010

Ans. (a)

See the explanation of above question.

150. When beams of red, blue and green lights fall on the same spot, the colour of the light becomes :

- (a) Violet
- (b) Red
- (c) Yellow
- (d) White

R.A.S./R.T.S. (Pre) 2003

Ans. (d)

Red, green and blue are referred to as the primary colours of light. Because we add them together to create a colour, we call them additive colors. Cyan, Magenta and yellow are called the subtractive color because we create colour by using them to subtract colour from white light. If we add all 3 primary colours of light in equal proportions we obtain white colour light.

151. Which is not a primary colour?

- (a) Black
- (b) Yellow
- (c) Red
- (d) Blue

M.P.P.C.S. (Pre) 2015

Ans. (a)

Black is not a primary colour. In painting red, yellow and blue are primary colours. But in terms of physics and light, primary colours are red, green and blue.

152. Three colours are the primary colours. These are -

- (a) Blue, Yellow and Red
- (b) Blue, Green and Red
- (c) Yellow, Green and Red
- (d) Blue, Yellow and Red

M.P.P.C.S. (Pre) 2005

Ans. (b)

See the explanation of above question.

153. Consider the following statements :

1. If magenta and yellow coloured circles intersect, the intersected area will have a red colour.
2. If cyan and magenta coloured circles intersect, the intersected area will have a blue colour.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

I.A.S. (Pre) 2007

Ans. (c)